

DA541A: Data Communication and Networks 7,5 credits

Lab manual 2

Purpose of the Lab

The purpose of this lab is to develop the student's ability to understand the principles behind switch operation, including its logic and address table, as well as to gain initial experience with Cisco's Command Line Interface (CLI) through basic configuration tasks. Additionally, the lab aims to build foundational skills in configuring both switches and routers, and to provide an understanding of the function and purpose of static and dynamic routing.

Litteraturhänvisning

Comer, Douglas E. (2014). *Computer Networks and Internets*, 6th Edition. Pearson.
Covers: Chapters 5–6, 9, 11, and 14

Equipment

You will find simulated equipment in Packet Tracer. In this lab students will be using
Cisco Switch 2960-24TT
Cisco Router 2901
PC as endpoint device

Note: The simulation tool used during the lab does not fully replicate the features of real physical network equipment. In a real-world scenario, access to network devices typically requires a console cable or a network connection via SSH.

1. Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation tool used for lab exercises. To install and use it:

1. Create a Cisco NetAcad Account:
 - Go to <https://www.netacad.com>.
 - Click on Sign up and register for a free account. You must enroll in a course to download Packet Tracer.
2. Enroll in a Course (Free):
 - After signing in, visit <https://www.netacad.com/courses/packet-tracer>.
 - Choose "*Introduction to Packet Tracer*" (free).
 - Enroll and start the course (you don't need to complete it to download the software).
3. Download Packet Tracer:
 - Once enrolled, go to *Resources > Download Packet Tracer*.
 - Select the appropriate version for your operating system (Windows/macOS/Linux).
4. Install the Program:
 - Run the downloaded installer and follow the instructions.
 - Log in using your Cisco NetAcad credentials.
5. Launch the Application:
 - After installation, open Packet Tracer.
 - You're ready to start working on this lab!

There are numerous high-quality tutorials available on how to use Packet Tracer. The link below directs you to Cisco's official website, where you can find comprehensive learning resources.

[Cisco Packet Tracer](#)

2. Switch configuration.

A switch is a plug-and-play device that functions without requiring any initial configuration. By default, it forwards Ethernet frames between connected devices, allowing them to communicate within the same local area network (LAN). This makes switches immediately operational in most basic networking environments.

2.1 Basic setup

Begin with adding a switch according to figure 1. Note that you need to enter the correct subcategory (green arrow). After that you can choose a component (blue arrow) You can drop and drag the components that you intend to use.

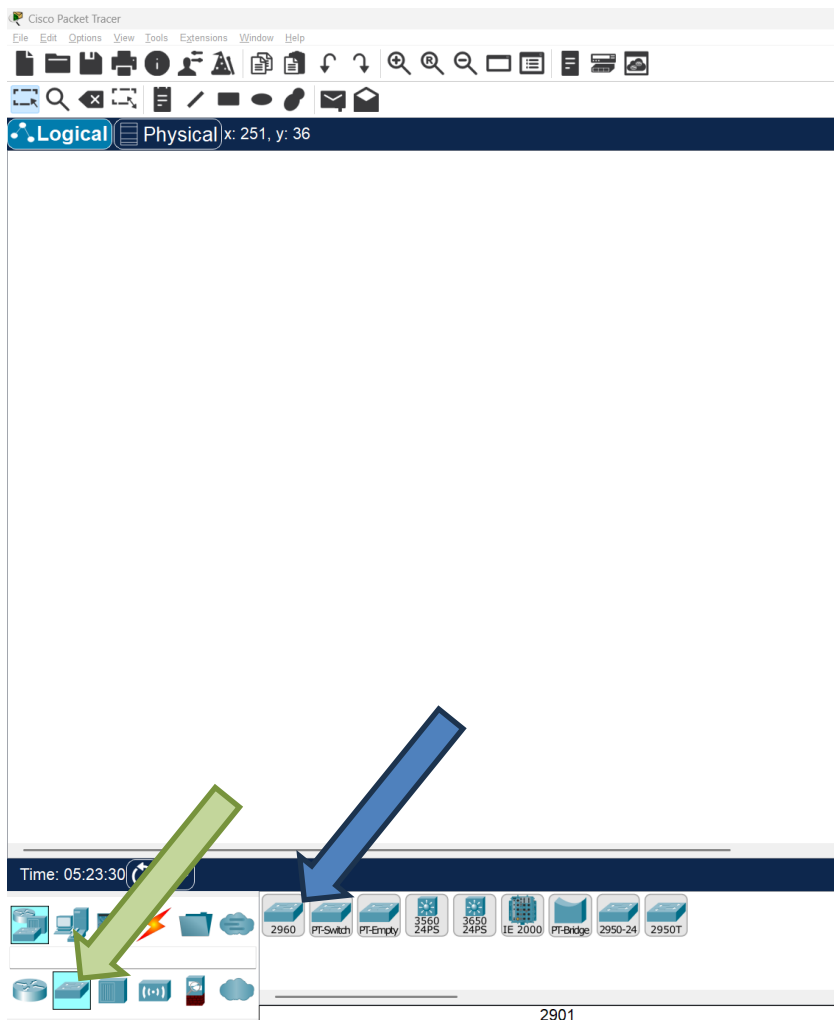


Figure 1

2.2 Information about the Command-Line Interface (CLI)

Cisco IOS Command Modes Overview

Cisco IOS devices, such as routers and switches, including those simulated in Packet Tracer operate through a structured command-line interface (CLI) organized into several command modes. The three fundamental modes are:

1. User EXEC Mode *hence forward referred to as **User Mode***

- Access: Automatically entered upon device login.
- Prompt: Router>
- Function: Allows limited, non-disruptive commands such as ping, show, and traceroute.
- Restrictions: Configuration commands are not available in this mode.

2. Privileged EXEC Mode *hence forward referred to as **Enable Mode***

- Access: Entered by typing enable from User EXEC Mode.
- Prompt: Router#
- Function: Grants access to all monitoring commands and administrative functions. You must be in this mode to enter configuration mode.

3. Global Configuration Mode *hence forward referred to as **Config Mode***

- Access: Entered using configure terminal from Privileged EXEC Mode.
- Prompt: Router(config)#
- Function: Enables you to make configuration changes to the device such as setting interface IP addresses, routing protocols, passwords, and more.

Official Cisco Documentation

You can find Cisco's formal explanation of these command modes and the CLI hierarchy here: [CISCO](#)

This page outlines the structure of command modes, including examples of how commands differ by context and privilege level.

The command mode that you are always taken to initially is User Mode, which is identified by the prompt `Switch>` (where the first part is the name of the device and the second part indicates the command mode).

To make it easier to navigate the wide range of commands available in each mode, Cisco has implemented a command help function, similar to the one found in the Windows command prompt. This help feature is accessed by typing a question mark (?).

For example: `Switch> ?`

When you enter a question mark, the possible commands or additional parameters will be displayed, for example, if you type `show ?`. Alternatively, you may see `<cr>` (Carriage Return), which means that the current input is sufficient to execute the command for instance, `show users ?`.

Exercise 1: Entering and Identifying Command Modes

1. Add a Cisco 2960 switch to the workspace in Packet Tracer.
2. Click the switch and open the CLI tab.
3. Identify which command mode appears by default upon startup.
4. Enter Enable Mode by typing: `Switch>enable`
5. Enter Global Configuration Mode: `Switch#configure terminal`
6. Go back to Enable mode by typing: `Switch(config)#exit`

Exercise 2: Basic configuration

Try to solve the tasks given below

1. Add Assign a **hostname** to the switch.
2. Set an **enable password**. Memorise the password outhewise you may lose access to the component.
3. Set the **clock**:
4. Save the configuration to the startup file. After you have done this restart the switch with the reload command and verify that your configuration is still active.
5. Erase the current configuration:
6. Restart the switch again
7. Verify that the switch has returned to default settings (e.g., `Switch>` prompt and no hostname).

Reflection and Review Questions

1. Which command mode do you enter first when launching the CLI? How do you recognize it?
2. What does the prompt (config)# indicate, and how do you reach this mode?
3. Why is it necessary to save the configuration to the startup file? What happens if you don't?
4. What information does the show running-config command display?
5. How can you confirm that a password has been applied? Can you view it in plain text?
6. What happens to the clock settings after a reboot if the switch has no battery backup?
7. How can you verify that the configuration was erased after using write erase and reload?

For the following questions you will need to use references, use the Vancouver reference system.

8. What are the potential security risks of not setting an enable password?
9. What is the difference between using the reload command and physically turning the switch off and on in Packet Tracer?
10. What is the difference between User EXEC Mode and Privileged EXEC (Enable) Mode?

2.3 Using Shortcuts and Auto-Completion in Cisco CLI

The Cisco IOS CLI allows you to type only part of a command as long as it is unambiguous and the switch will recognize it.

For example, instead of typing the full command:

```
configure terminal
```

You can simply type:

```
conf t
```

and it will still work the same way.

You can also press the Tab key after typing part of a command to automatically complete it.

For example:

- Typing `sh ru` and then pressing Tab will complete it to show running-config (if there are no other similar commands).
- Typing `en` and pressing Tab will complete it to `enable`.

If there are multiple commands that begin the same way, you will see a list of suggestions instead.

This feature helps you:

- Save time
- Avoid typos
- Discover available commands

Try using the Tab key and abbreviations throughout the exercises!

3 Lab: MAC Tables

In this exercise, you will examine the MAC tables in a switch and also observe the difference between clients connected to the same switch versus clients connected to different switches. Remember that you also need to configure an IP address on your clients. In this lab, we are using the network 192.168.1.0 with subnet mask 255.255.255.0. you should also identify the MAC addresses of your clients.

Task 1

Connect two clients to the same switch, as shown in figure 2 . Make sure they can communicate with each other by using the “ping” command. Once everything is working and the clients can reach each other, return to the switch.

The **ping** command is a simple network tool used to test connectivity between two devices. It works by sending **ICMP (Internet Control Message Protocol) Echo Request** messages to a target host and waiting for an **ICMP Echo Reply**.

If replies are received, it confirms that the devices can reach each other over the network. If not, it indicates that there may be a connectivity issue, such as a wrong IP address, a cable problem, or a firewall blocking ICMP traffic.

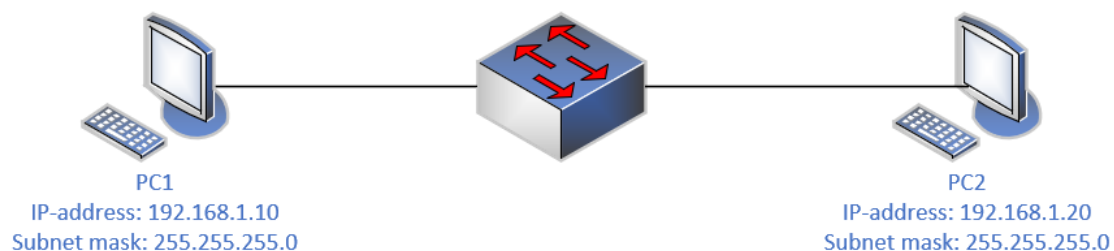


Figure 2

In order for a switch to know which interface an Ethernet frame should be forwarded to, it uses what is commonly called a **MAC table**. The switch learns this table automatically as traffic is generated in the network.

To view the MAC table, you only need to be in **user mode**. Simply type:

```
show mac address-table dynamic
```

Look at the MAC table and compare it with the MAC addresses of your clients and the interfaces they are connected to.

Task 2

Now connect two clients to separate switches, as shown in Figure 3. Ensure that they can communicate with each other by using the “ping” command.

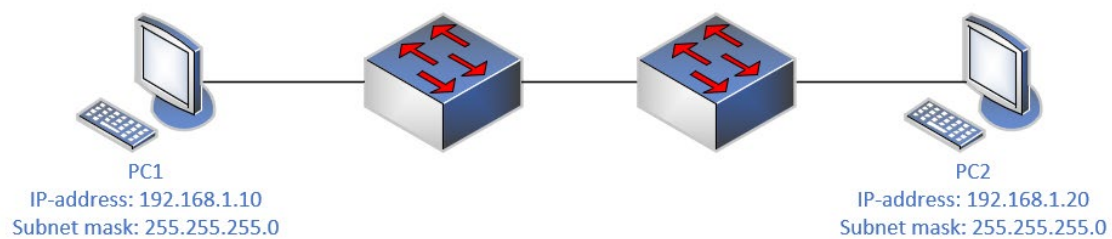


Figure 3

When everything is working and the clients can reach each other, log in to the switches and examine the MAC tables. Compare the MAC addresses of your clients with the interfaces they are connected to.

How do the MAC tables in this task differ from the previous task, where only a single switch was used?

Reflection and Review Questions

1. What information does the MAC table provide about the devices connected to the switch?
2. Why does a switch need a MAC table in order to forward Ethernet frames correctly?
3. How does the switch learn the MAC addresses of connected devices?
4. In Task 1, when both clients were connected to the same switch, what did you observe in the MAC table?
5. In Task 2, when the clients were connected to different switches, how did the MAC tables differ compared to Task 1?
6. Why is the **ping** command useful in this lab, and what does it tell you about the network connection?
7. What kind of network issues could cause a ping to fail, even if the devices are physically connected?

8. How does the concept of MAC tables relate to the difference between a **switch** and a **hub**?

4. Log in with Telnet

In a real network environment, you must use a console connection to access and configure a switch or router for the first time. This is because the device does not have an IP address or remote access enabled by default, so you cannot connect to it over the network initially. However, in Packet Tracer, this step is simplified. You can open the device directly and start configuring it without setting up a console connection first. This makes it easier to focus on learning the commands and network concepts rather than the physical setup.

One way to make configuration easier when dealing with real physical equipment is to set up the switch so that you can connect to it through a regular interface using the Telnet protocol.

In this lab, we will use Telnet even though it is extremely insecure and should always be replaced by SSH in real environments. However, since the configuration for Telnet is simpler and also forms the foundation for SSH, we will start with Telnet. In addition, our switches are not connected to the Internet, which reduces the risk of attacks.

To configure the switch for Telnet access, two steps are required. First, we need to set a password for remote access. Then, we must assign an IP address to the switch's built-in management interface.

Note: In real networks, **SSH** should always be used instead of Telnet, since Telnet transmits all data, including passwords, in plain text.

Steps to Enable Telnet Access on a Switch

1. Enter privileged EXEC mode

```
Switch> enable
```
2. Enter global configuration mode

```
Switch# configure terminal
```
3. Set a password for remote (vty) access

```
Switch(config)# line vty 0 4
Switch(config-line)# password cisco
Switch(config-line)# login
Switch(config-line)# exit
```
4. Set an enable password (to access privileged mode remotely)

```
Switch(config)# enable password class
```

5. Assign an IP address to the management interface (usually VLAN 1)

```
Switch(config)# interface vlan 1
Switch(config-if)# ip address 192.168.1.2 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

6. Configure a default gateway (if switch needs to be reached from another network)

```
Switch(config)# ip default-gateway 192.168.1.1
```

Now try logging in to the switch using Telnet. Just like with serial communication, you can use the Terminal. To connect, type:

```
telnet <switch IP-address>
```

That is, the IP address you assigned to the switch, followed by the username and password.

Note that you still need to enter your enable password if you want to move into enable mode, which provides an extra layer of security.

VLAN 1

By default, all switch ports belong to VLAN 1, which is the management VLAN on most Cisco switches. When we assign an IP address to VLAN 1, it allows the switch itself (not just the devices connected to it) to communicate on the network and be managed remotely.

Default-gateway

sets the switch's default gateway, which is the device used to reach networks outside of the local subnet. This is **not necessary in this lab**, since we are only working within a single local network (subnet). In real networks, however, a default gateway is essential, as it allows the switch to communicate with devices outside its own network, such as on other subnets or across the Internet.

Step 2: Assign an IP address to the switch. Normally, a switch does not have an IP address. However, you can assign an IP address to the internal processor connected to a virtual LAN (VLAN). For this lab, we will use VLAN 1 (the default VLAN).

From the CLI, enter configuration mode and type:

- `interface vlan 1`
- `ip address <address> <mask>`

Now, try logging in to the switch via Telnet. Just as with serial communication, you can use the terminal application. To connect, type ``telnet <switch address>``—using the IP address you assigned to the switch—and then enter the username and password.

Note that you still need to enter your enable password if you wish to switch to enable mode, providing an extra layer of security.

5. Configure interface

The next task is to configure the interfaces that your clients are connected to.

Assume that there is an interface on the switch labeled GigabitEthernet 0/1 or FastEthernet 0/1 that you want to configure.

To enter the configuration mode for this interface, type:

```
Switch# conf t
Switch(config)#interface GigabitEthernet 0/1
```

Note that abbreviations are commonly used to quickly enter configuration modes.

For example, instead of typing the full command, you can simply write: `int gi 0/1` to access the interface configuration mode.

You can also configure multiple interfaces at the same time by using the **range** command.

This is especially useful if several ports should have the same configuration.

For example, on a Cisco 2960 switch you might type:

```
Switch(config)# interface range fastEthernet 0/1 - 12
```

This allows you to apply the same settings to all interfaces from FastEthernet 0/1 through FastEthernet 0/12 at once.

Configuring Port Security with Sticky MAC Addresses

Another useful feature is called Port Security. Port Security is used to control which MAC addresses are allowed on a specific interface. You can either manually specify the permitted MAC addresses, or you can let the switch automatically limit access for new clients using a method called sticky.

With sticky MAC addresses, the switch automatically learns and registers the MAC address of the device connected to the interface. To configure a sticky MAC address, follow these steps:

1. Select the interface

Enter the configuration mode for the interface you want to secure:

```
Switch(config)# interface fastEthernet 0/1
```

2. Enable port security

Activate port security on the selected interface:

```
Switch(config-if)# switchport port-security
```

3. Set maximum number of MAC addresses

Define how many MAC addresses are allowed on this port (commonly 1 for single-client ports):

```
Switch(config-if)# switchport port-security maximum 1
```

4. Enable sticky MAC learning

Enable the switch to automatically learn and save the first MAC address that connects to this interface:

```
Switch(config-if)# switchport port-security mac-address sticky
```

After this, the learned MAC address will appear in the running configuration and be treated as if it was manually configured.

5. Define violation mode (optional but recommended)

Choose the action if an unauthorized device tries to connect:

- protect → Drops traffic from unauthorized MACs silently.
- restrict → Drops traffic and logs a violation.
- shutdown → Disables the port (default).

Example:

```
Switch(config-if)# switchport port-security violation restrict
```

6. Exit and save configuration

```
Switch(config-if)# exit
```

```
Switch# write memory
```

Verifying Sticky MAC Addresses

To check which MAC addresses are learned and secured by sticky:

```
Switch# show port-security interface fastEthernet 0/1
```

To see sticky MAC addresses specifically:

```
Switch# show running-config
```

Here, you will see the automatically learned MAC address listed under the interface configuration as a *sticky* address.